Lasso Trading

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1 Introduction

The volume and granularity of data available worldwide is growing at an unprecedented pace. In 2024 alone, the world generated approximately 402 million terabytes of data per day, up from 43 million just a decade earlier (Wright, 2025). Financial markets are no exception to this trend. Whereas investors once had access to only a handful of data series, today they draw on a vast array of signals, ranging from traditional balance-sheet information to textual data, satellite imagery, and web traffic. For most of these new variables, there is little theoretical guidance about whether they should forecast stock returns. Nevertheless, demand for alternative data is strong, and investors increasingly use machine learning and artificial intelligence (AI) to search for predictive patterns in large datasets (Invesco, 2024). These methods require no theoretical guidance about which variables should matter; instead, they let the data speak for themselves.

This behavior is not well captured by existing asset pricing models with learning. In these models, agents typically are assumed to employ self-referential or correctly specified forecasting rules, or to work with severely under-parameterized perceived laws of motion. Agents track a small number of parameters and update them recursively using variants of least squares or constant-gain learning. While analytically convenient, these frameworks do not describe the behavior of modern investors who consider thousands of potential signals to forecast returns and use flexible statistical procedures to discover relationships without an explicit structural model in mind.

We develop a model that embeds this form of statistical learning into a standard asset pricing environment. Agents suspect that returns may be conditionally predictable from a large set of observable predictors. To exploit this perceived opportunity, they bet on sparsity and apply the apply the least absolute shrinkage and selection operator (LASSO) to select a sparse subset of variables from the high-dimensional predictor space (Tibshirani, 1996).

Our main result is that the very act of searching for predictability with high-dimensional statistical learning can generate short-lived pockets of return predictability. In equilibrium, these pockets are sparse and evolve over time, echoing recent evidence on episodic short-horizon predictability (Chinco, Clark-Joseph and Ye, 2019; Farmer, Schmidt and Timmermann, 2023). The mechanism follows a simple life cycle. In finite samples, LASSO occasionally selects predictors whose estimated coefficients are large enough to survive the penalty. This concentrates attention on a small number of apparently strong relationships while shrinking most coefficients to zero. When investors trade on the resulting perceived predictability, their demand feeds back into prices, turning the perceived signal into genuine but temporary predictability. As new data arrive and old observations are discounted, the selected coefficients shrink back toward zero, the signal disappears, and prices revert to behavior consistent with unconditional unpredictability.

We reach this result, by formalizing this mechanism in a Lucas-type exchange economy with a single risky asset. Under rational expectations, the price–dividend ratio is constant and returns are unpredictable. We follow Adam, Marcet and Nicolini (2016) and depart from rational expectations by assuming that agents know the dividend and consumption process but not the process for returns. Instead, the agents suspect that returns are conditionally predictable from a large set of observable signals. Each period, investors estimate a perceived law of motion (PLM) in which next periods return depends linearly on a sparse subset of predictors, that the agents selectes using LASSO. As the number of candidate predictors exceeds the sample size by a large margin, standard linear estimation techniques are infeasible. Beliefs then feed into equilibrium prices through the actual law of motion (ALM) governing returns. In contrast to the linear PLM, the ALM is nonlinear and thus misspecified. In this case, learning is unlikely to converge to the true law of motion. Instead, the economy converges to a cross-moment consistent equilibrium (CMCE) in which agents estimated forecasting moments coincide with those generated by the equilibrium itself. Occasional variable-selection events can then trigger predictable return episodes that persist only until the corresponding coefficients shrink back toward zero.

We test the models predictions using the cross-section of US stocks and a set of 180 news topics constrcuted in Bybee et al. (2021) using the Wall Street Journal. Our model is able to [....]

The rest of the paper is structured as follows. Section 2 discusses related literature. Section 3 presents the model environment. Section 4 analyzes the asset-pricing equilibrium under standard Recursive Least Squares. Section 5 examines Constant Gain Learning. Section 6 introduces LASSO learning and derives our main results regarding sparse predictability. Section 7 estimates the model on the cross-section of US stocks using a high-dimensional set of textual predictors. Section 8 concludes.

2 Related Literature

Our model relates to the large literature on asset pricing with learning. The central premise of this work is that investors have incomplete knowledge of the data-generating process governing market outcomes and therefore infer it from observed data. Learning is often introduced to account for empirical features, such as excess volatility and return predictability, that are difficult to replicate with fully informed agents.

Early contributions assume fully rational agents who estimate unknown parameters of a correctly specified model using least squares (Marcet and Sargent, 1989). In this setting, learning converges to the rational expectations equilibrium (REE) as agents observe more data. Even in this low-dimensional environment, learning can generate excess volatility and return predictability (Timmermann, 1993, 1996). Convergence to the REE is no longer guaranteed when agents are boundedly rational; with finite-memory and finite-sample learning, beliefs may fail to converge (Honkapohja and Mitra, 2003).

The implications become richer when agents face model uncertainty in addition to parameter uncertainty. Barberis, Shleifer and Vishny (1998) study investors who switch between competing forecasting rules, while Hong, Stein and Yu (2007) and Branch and Evans (2010) analyze settings in which agents estimate univariate perceived laws of motion despite a multivariate true model.

More recently, Adam and Marcet (2011) introduce the concept of *internal rationality*, where agents optimize decisions given subjective beliefs even if those beliefs are misspecified. Building on this, Adam, Marcet and Nicolini (2016) embed internally rational agents in a standard asset-pricing model in which investors estimate the mapping from fundamentals to prices, generating a feedback loop that helps match key asset-pricing moments. Our model builds on Adam, Marcet and Nicolini (2016) by adopting their general equilibrium structure, but replaces their learning rule with a microfounded LASSO-based mechanism. This distinction is central: whereas their framework generates persistent belief-driven dynamics, our variable-selection mechanism delivers *episodic* and *sparse* predictability, consistent with evidence from high-dimensional return prediction.

Our paper also relates to research on how big data and improved statistical tools affect asset prices. Begenau, Farboodi and Veldkamp (2018) show that higher data-processing capacity improves investor forecasts and changes equilibrium prices and the cost of capital, while Farboodi and Veldkamp (2020) model investors who optimally choose whether to acquire data about fundamentals or about the behavior of other investors.

Empirical evidence indicates that return predictability is time-varying. Variables such as consumption growth and valuation ratios forecast returns in some periods but not in others (**???**). A complementary strand emphasizes sparsity: among the many predictors proposed in the literature, only a small and time-varying subset appears informative at any point in time (Kozak, Nagel and Santosh, 2020; Freyberger, Neuhierl and Weber, 2020). Mclean and Pontiff (2016) show that many return anomalies attenuate after publication, consistent with investors actively searching for and trading on these signals.

Machine learning techniques are increasingly used to handle the ever-growing number of return predictors and to mitigate data-mining concerns (Feng, Giglio and Xiu, 2020). Gu, Kelly and Xiu (2020) demonstrate that machine-learning forecasts outperform traditional linear models by exploiting sparse signals and nonlinear interactions. Most closely related to our learning mechanism is Chinco, Clark-Joseph and Ye (2019), who document that LASSO uncovers short-lived pockets of high-frequency predictability concentrated in a small, shifting set of predictors. Finally, our work connects to initial attempts to model market outcomes when agents use statistical learning. Georges and Pereira (2021) study adaptive model selection and regularization in an agent-based market simulation. Relative to this simulation-based evidence, our contribution is to provide an analytical characterization of LASSO-based learning in an equilibrium environment and to derive sharp conditions under which sparse, temporary predictability arises endogenously.

3 The model

We study a standard Lucas (1978) asset pricing model in the flavour of Adam, Marcet and Nicolini (2016). We consider an economy populated by a continuum of agents of mass one who trade a single risky asset. The asset pays a stochastic dividend D_t at each discrete time period $t = 0, 1, 2, \ldots$. The dividend and consumption processes are driven by multivariate normally distributed shocks

$$\begin{pmatrix} \eta_t^d \\ \eta_t^c \end{pmatrix} = \begin{pmatrix} \log \varepsilon_t^d \\ \log \varepsilon_t^c \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} -\sigma_d^2/2 \\ -\sigma_c^2/2 \end{pmatrix}, \begin{pmatrix} \sigma_d^2 & \rho\sigma_d\sigma_c \\ \rho\sigma_d\sigma_c & \sigma_c^2 \end{pmatrix} \right), \tag{1}$$

The dividend process follows

$$\frac{D_t}{D_{t-1}} = a\varepsilon_t^d, \tag{2}$$

where $a \geq 1$ is a constant growth factor, which implies that

$$\mathbb{E}_t\left[D_{t+1}\right] = aD_t. \tag{3}$$

Similarly for the aggregate consumption process we postulate that

$$\frac{C_t}{C_{t-1}} = a\varepsilon_t^c. \tag{4}$$

The equilibrium price of the asset is determined by the standard no-arbitrage condition which requires that the price at time t equals the discounted expected value of next period's price plus dividend, that is

$$P_t = \delta\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\ell} P_{t+1} + D_{t+1} \right], \tag{5}$$

where δ is the discount factor, ℓ is the risk aversion parameter[1], and $\mathbb{E}_t[\cdot]$ is the expectation operator conditional on information available at time t. Under FIRE, the equilibrium price can be solved as

$$P_t = \frac{\delta a^{(1-\ell)}\rho_e}{1 - a^{(1-\ell)}\rho_e} D_t, \tag{6}$$

where

$$\rho_e = \mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\ell} \frac{D_{t+1}}{D_t} \right] = e^{\ell(1+\ell)\sigma_c^2/2} e^{-\ell\rho\sigma_c\sigma_d}. \tag{7}$$

[1]The consumption based asset pricing literature normally used γ, but we will use ℓ to avoid confusion with the learning gain parameter

This implies a constant price-dividend ratio and an ex-dividend gross return which is given by

$$R_{t+1} = \frac{P_{t+1}}{P_t} = \frac{D_{t+1}}{D_t} = a\varepsilon_{t+1}^d. \tag{8}$$

Taking logs we obtain log-returns

$$r_{t+1} = \log(R_{t+1}) = \log(a) + \eta_{t+1}^d \sim \mathcal{N}(\log(a) - \sigma_d^2/2, \sigma_d^2). \tag{9}$$

Without loss of generality in what follows we set $\log(a) = \sigma_d^2/2$. This implies that under FIRE log-returns have mean zero and there is no time-t information that can help forecasting returns.

Instead of assuming that agents know the true data generating process, we follow the adaptive learning literature and assume that agents are boundedly rational and form their expectations using econometric techniques [citations here]. Equation (5) shows that agents have to forecast two quantities, the risk-adjusted gross return and gross dividend growth. We follow Adam, Marcet and Nicolini (2016) in making the next assumption.

Assumption 1 *Agents know the process for dividends and consumption and are boundedly rational only with respect to (gross) returns.*

In this way we can isolate the effects of learning about returns on asset prices.

3.1 Univariate PLM and ALM

To fix ideas we start by considering the univariate case in which agents try to predict log-returns using a single predictor x_t. Specifically we assume that agents have a linear and stationary PLM for the asset return

$$r_{t+1} = x_t\beta + u_{t+1}, \tag{10}$$

with $x_t \sim \mathcal{N}(0, \sigma_x^2)$ being a signal available for the agent at time t which they suspect might have predictive power on log-returns. Then the agents' PLM about log-returns implies the following PLM for gross returns

$$R_{t+1} = e^{x_t\beta + u_{t+1}}. \tag{11}$$

This implies that gross expected[2] returns follow a log-normal distribution and their conditional forecast is

$$\tilde{\mathbb{E}}_t(R_{t+1}) = \tilde{\mathbb{E}}_t\left(\frac{P_{t+1}}{P_t}\right) = e^{x_t\beta + \sigma_u^2/2} = \phi e^{x_t\beta}, \tag{12}$$

by letting $\phi = e^{\sigma_u^2/2}$.

[2]Throughout the paper we use $\tilde{\mathbb{E}}_t[\cdot]$ to denote expectations taken under the agents' PLM, while $\mathbb{E}_t[\cdot]$ denotes rational expectations.

Assumption 2 *Within their perceived law of motion, agents believe that the return process and consumption growth are independent.*

Assumption 2 allows us to derive risk-adjusted expected gross returns as

$$\tilde{\mathbb{E}}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\ell}\frac{P_{t+1}}{P_t}\right]=\tilde{\mathbb{E}}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\ell}\right]\tilde{\mathbb{E}}_t\left(\frac{P_{t+1}}{P_t}\right)=a^{-\ell}\phi e^{x_t\beta}. \tag{13}$$

Plugging this into (5) yields

$$P_t=\frac{\delta a^{(1-\ell)}\rho_e}{1-\delta a^{-\ell}\phi e^{x_t\beta}}D_t. \tag{14}$$

Now divide by P_{t-1} to obtain actual gross returns

$$R_t=\frac{P_t}{P_{t-1}}=\frac{1-\delta a^{-\ell}\phi e^{x_{t-1}\beta}}{1-\delta a^{-\ell}\phi e^{x_t\beta}}\frac{D_t}{D_{t-1}}, \tag{15}$$

and taking logs and moving one step forward we obtain the ALM

$$r_{t+1}=\eta_{t+1}+\log(1-\kappa e^{x_t\beta})-\log(1-\kappa e^{x_{t+1}\beta}), \tag{16}$$

where $\kappa:=\delta a^{-\ell}\phi$ and $\eta_{t+1}=\log(a)+\eta^d_{t+1}$.

Expression (16) is only well-defined when $1-\kappa e^{x_t\beta}>0$ and $1-\kappa e^{x_{t+1}\beta}>0$ hold with probability one, which imposes state-dependent restrictions on (x_t,β). To avoid these domain issues and guarantee global existence of a stochastic equilibrium, we adopt a bounded specification for the conditional mean of returns. We modify the perceived conditional expectation of gross returns to

$$\tilde{\mathbb{E}}_t(R_{t+1})=\phi\exp(w\tanh(x_t\beta/w)), \tag{17}$$

where $w=-\log(\kappa)-\varepsilon$ for some $\varepsilon>0$. Since $\tanh(\cdot)\in(-1,1)$, we have $w\tanh(x_t\beta/w)\in(-w,w)$ which implies $e^{w\tanh(x_t\beta/w)}\in(e^{-w},e^w)$. By construction $w<-\log(\kappa)$, so $e^w<\kappa^{-1}$ and therefore $1-\kappa e^{w\tanh(x_t\beta/w)}>0$ for all x_t and β.

The corresponding risk-adjusted expectations and price-dividend ratio are defined as before, with $x_t\beta$ replaced by $w\tanh(x_t\beta/w)$. This yields the modified ALM

$$r_{t+1}=\eta_{t+1}+\log(1-\kappa e^{w\tanh(x_t\beta/w)})-\log(1-\kappa e^{w\tanh(x_{t+1}\beta/w)}). \tag{18}$$

For later use, we define the function

$$g_\beta(x):=\log(1-\kappa\, e^{w\tanh(x\beta/w)}), \tag{19}$$

which is globally well-defined and smooth in x and β. the original ALM (16) Given that the term $x\beta$ represents expected returns, we argue that the original ALM is well-defined with high probability. The bounded specification (18), should therefore be interpreted as a globally valid extension that coincides with the baseline model in the empirically relevant region of the state space while eliminating corner cases where the logarithm is undefined. As noted above, agents do not know the value of β in their PLM (10) and have to estimate it from observed data. We therefore consider different learning mechaisms that agents can use to update their beliefs about β over time, sequentially building towards the LASSO learning mechanism that is the main focus of this paper.

4 Recursive Least Squares Learning

We first consider Recursive Least Squares (RLS) learning, which is a recursive implementation of standard Ordinary Least Squares (OLS). At each time t, agents update their estimate β_t by minimizing the sum of squared forecast errors:[3]

$$\beta_t = \left(\sum_{i=1}^{t-1} x_{i-1}^2 \right)^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{20}$$

This estimator admits a recursive formulation. Define $M_{t-1} = \frac{1}{t-1} \sum_{i=1}^{t-1} x_{i-1}^2$. Standard algebra, which we defer to Appendix (A.1) yields the recursions

$$M_t = M_{t-1} + \frac{1}{t} \left(x_{t-1}^2 - M_{t-1} \right), \tag{21}$$

$$\beta_t = \beta_{t-1} + ((t-1)M_{t-1})^{-1} \left(x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-1} \right). \tag{22}$$

Under RLS learning, actual returns evolve according to

$$r_{t-1} = \eta_{t-1} + g_{\beta_{t-2}}(x_{t-2}) - g_{\beta_{t-1}}(x_{t-1}), \tag{23}$$

where $g_\beta(\cdot)$ is defined in (19).

A natural question is whether this learning process converges to the rational expectations equilibrium. Under full information rational expectations (FIRE), returns in our model are unpredictable and $\beta = 0$. However, convergence to the REE requires not only that agents' perceived relationship matches the actual relationship, but that the entire distributions implied by the PLM and ALM coincide. When the ALM is nonlinear and the PLM is linear, such convergence is generally not possible. The learning literature has instead focused on weaker notions of equilibrium that require consistency only on specific moments rather than on entire distributions (Hommes and Zhu, 2014). Following this approach, we define a Cross-Moment Consistent Equilibrium (CMCE) as an equilibrium in which the population regression coefficient in agents' perceived linear model matches the projection of actual returns onto the predictor.

Definition 1 (Cross-Moment Consistent Equilibrium) *A pair (μ, β) is a* Cross-Moment Consistent Equilibrium (CMCE) *if:*

[3]We use the following timing: at time t the agent forecasts $r^e_{t+1} = x_t \beta_t$ which then implies a realization for P_t and subsequently r_{t+1}. Therefore, to avoid simultaneity issues, we assume that when computing the current estimate, the last available log-return observation is r_{t-1}.

1. *μ is a non-degenerate invariant probability measure for the stochastic process*

$$r_{t-1} = \eta_{t-1} + g_\beta(x_{t-2}) - g_\beta(x_{t-1}),$$

where $x_t \overset{iid}{\sim} \mathcal{N}(0, \sigma_x^2)$.

2. *The parameter β is the population OLS coefficient of r_{t-1} on x_{t-2} under the invariant measure μ, that is,*

$$\beta = \frac{\mathrm{Cov}_\mu(x_{t-2} r_{t-1})}{\mathrm{Var}_\mu(x_{t-2}^2)}$$

While a rational expectations equilibrium requires that agents' perceived law of motion coincides with the actual law of motion, a CMCE requires only that the population regression coefficient in agents' perceived linear model matches the projection of actual returns onto the predictor x_{t-2}. This weaker notion of consistency is appropriate when agents use misspecified linear models to forecast returns generated by a nonlinear process. Despite the misspecification, there is a statistical self-consistency between the empirical relationship agents perceive and the one implied by the ALM.

Proposition 1 *Under RLS learning there exists a unique CMCE given by $\beta = 0$. The CMCE is locally stable under learning for $\kappa \in (0, 1)$.*

Proof. In Appendix B.1.

Notice that $\kappa = \delta a^{-\ell} \phi$ is strictly positive for all admissible parameter values. The restriction $\kappa < 1$ imposes an upper bound on the combination of the discount factor, risk aversion, and dividend growth, and is required for the REE to be well-defined. This condition holds for standard parameter calibrations.

Proposition 1 shows that with infinite memory, learning eliminates perceived predictability: agents eventually accumulate enough data to recognize that returns are fundamentally unpredictable. However, this convergence relies critically on the assumption that all historical observations receive equal weight in estimation. In practice, investors may discount older data either because they believe the economic environment is changing or because they face cognitive or computational constraints on memory. We now examine how limited memory alters the learning outcome.

5 Weighted Least Squares and Constant Gain Learning

We saw that memory, in the sense of the weight assigned to past observations, plays a key role in determining the learning outcome. Under RLS learning with decreasing gain, agents eventually learn that returns are unpredictable. However, in practice agents may assign more weight to recent observations, either because they believe that the data-generating process is time-varying, or because they have limited memory capacity. To capture this, the second learning mechanism we consider is constant gain learning. This obtains by replacing the decreasing gain $1/t$ in RLS with a constant gain $g \in (0, 1)$, and as we show in Appendix A.2 constant gain corresponds to an on-line estimation of WLS (Weighted Least Squares) with geometrically declining weights.

$$\beta_t = \beta_{t-1} + \gamma M_{t-1}^{-1} x_{t-2}(r_{t-1} - x_{t-2}\beta_{t-1}). \tag{24}$$

$$M_t = M_{t-1} + \gamma \left(x_{t-1}^2 - M_{t-1} \right). \tag{25}$$

$$r_{t-1} = \eta_{t-1} + \log\Big(1 - \kappa e^{w \tanh(x_{t-2}\beta_{t-2}/w)}\Big) - \log\Big(1 - \kappa e^{w \tanh(x_{t-1}\beta_{t-1}/w)}\Big). \tag{26}$$

Given that the gain is not decreasing and that the updating depends on the stochastic term η_{t-1}, differently to RLS, the learning mechanism cannot converge to a fixed point. However, under suitable conditions and for small values of γ, and under the same stability condition $\kappa \in (0,1)$, it can be shown that the learning dynamics will fluctuate around the CMCE. We formalize this in the following proposition.

Proposition 2 (Distribution under Constant Gain Learning) *Under Constant gain learning withwith parameter γ, and for $\kappa < 1$, for small values of γ the estimated parameter β approximately follows a normal distribution*

$$\beta_t \sim \mathcal{N}\left(\beta^*, \gamma \frac{\sigma_d^2(1-\kappa)}{2\sigma_x^2} \right), \tag{27}$$

where $\beta^* = 0$ is the cross-moment consistent equilibrium.

Proof. In Appendix B.2.

Proposition 2 characterizes the stochastic fluctuations in beliefs under finite memory. The variance of the estimated coefficient, $\gamma\sigma_d^2(2\sigma_x^2(1-\kappa))^{-1}$, is driven by three fundamental parameters.

First, the gain parameter γ controls memory length and directly amplifies variance. Higher γ means agents discount past data more aggressively, giving recent observations disproportionate influence. This amplifies the impact of sampling noise: a few unusual return realizations can substantially shift beliefs when historical evidence receives little weight.

Second, dividend volatility σ_d^2 determines the fundamental uncertainty in returns. Greater return volatility generates larger forecast errors, which translate into more volatile coefficient estimates as agents attempt to fit a predictive relationship to noisy data.

Third, predictor variance σ_x^2 acts as a stabilizing force. When the predictor varies widely, each observation provides more information about the forecasting relationship, reducing estimation uncertainty.

6 Lasso Regression

We model sparse belief formation by endowing agents with a *soft-thresholded least squares* learning rule. Concretely, agents first update a least squares estimate of the forecasting coefficient using recursive or constant-gain learning, and then apply a soft-thresholding operator before forming expectations. This mechanism preserves analytical tractability while capturing the key feature of penalized learning, namely the tendency to set coefficients exactly to zero when empirical signals are weak.

Our empirical motivation is closely related to LASSO regression, which augments the least squares objective with an ℓ_1 penalty and is widely used to promote sparsity in estimated models. Rather than solving the LASSO problem directly within the learning dynamics, we adopt the soft-thresholding rule as a reduced-form representation of penalized estimation. As shown in Appendix A.3, this procedure is exactly equivalent to LASSO in the univariate case when regressors are orthonormal, in which case the LASSO estimator admits the closed-form representation

$$\hat{\beta}_{\text{LASSO}} = S(\beta_{\text{OLS}}, \lambda) = \text{sign}(\beta_{\text{OLS}}) \max\{0, |\beta_{\text{OLS}}| - \lambda\} . \tag{28}$$

While exact orthonormality is unlikely to hold in practice, it is common to work with predictors that are uncorrelated and standardized. In such environments, soft-thresholding provides a close approximation to the LASSO solution and is routinely used in empirical applications. We therefore refer to this learning mechanism as *LASSO learning*, with the understanding that it implements penalized least squares through an analytically tractable thresholding rule.

Under this specification, learning dynamics can be studied using the same recursive framework developed for least squares learning. The unpenalized coefficient continues to satisfy the standard cross-moment consistency condition, while sparsity arises endogenously from the thresholding step. As we show below, this modification leaves the location of the cross-moment consistent equilibrium unchanged but alters the stochastic behavior of beliefs under constant-gain learning. Specifically, the estimate for β_{OLS} will be distributed as a normal distribution centered around the fixed point, with variance proportional to the gain parameter γ. This implies that for small λ sometimes the estimates will be large enough that the LASSO estimate is non-zero. Given the distribution of β_{OLS}, we have:

Proposition 3 (Probability of Non-Zero LASSO Estimate) *Under Weighted Least Squares learning with geometrically declining weights with parameter γ, and for $\kappa < 1$, for small values of γ the probability that the LASSO estimate is different from zero is given by*

$$\mathbb{P}(|\beta_{OLS}| > \lambda) = 1 - \left[\Phi\left(\frac{\lambda}{\sigma_{OLS}}\right) - \Phi\left(\frac{-\lambda}{\sigma_{OLS}}\right)\right], \tag{29}$$

where $\sigma_{OLS} = \left(\gamma s_d^2 \left(2\sigma_x^2 \left(1 + \frac{\kappa}{1-\kappa}\right)\right)^{-1}\right)^{1/2}$.

Proof. In Appendix B.3.

It is worth claryifing now the mechanism through which return predictability arises in our model. Under constant-gain learning, the evolution of beliefs can be written in on-line form as

$$\beta_t = \beta_{t-1} + \gamma M_{t-1}^{-1} x_{t-2}(r_{t-1} - x_{t-2}\beta_{t-1}), \tag{30}$$

where the term $r_{t-1} - x_{t-2}\beta_{t-1}$ is the one-step-ahead forecasting error. Large changes in β_t therefore arise when the forecasting error is large, when the regressor realization $|x_{t-2}|$ is large, or when the local second-moment estimate M_{t-1} is small. Given that returns evolve according to the ALM (26), it is clear that even when predictors x and innovations η are i.i.d. and independent, realizations in which x_{t-2} and η_{t-1} are accidentally aligned in sign and magnitude generate large forecast errors and hence large belief revisions. Economically, this implies that the probability of a regressor being selected as predictive, is larger when it seems to agents that innovations to the fundamental dividend process are correlated with movements in the predictor.

6.1 Extension to Multiple Uncorrelated Predictors

Our theoretical analysis focuses on the univariate case for analytical tractability, but all main results extend naturally to the multivariate setting when predictors are uncorrelated. Consider a PLM with k predictors: $r_{t+1} = X_t^T\beta + u_{t+1}$, where $X_t \sim \mathcal{N}(0, \Sigma_X)$ and Σ_X is diagonal. In this case, the second-moment matrix $M_t = \mathbb{E}[X_t X_t^T]$ is also diagonal, which implies that the learning dynamics decompose into k independent univariate problems. Specifically, each coefficient β_j evolves according to

$$\beta_{j,t} = \beta_{j,t-1} + \gamma M_{j,t-1}^{-1} x_{j,t-2} \left(r_{t-1} - X_{t-2}^T \beta_{t-1} \right), \tag{31}$$

where $M_{j,t} = \mathbb{E}[x_{j,t}^2]$ is the j-th diagonal element. The cross-moment consistent equilibrium condition similarly decomposes: $\beta_j = T_j(\beta)$ for all j, where $T_j(\beta) = \mathrm{Cov}(r_{t-1}, x_{j,t-2})/\mathrm{Var}(x_{j,t-2})$. Under constant-gain learning with penalty parameter λ, each unpenalized coefficient is approximately normally distributed with mean zero and variance $\gamma\sigma_d^2/(2\sigma_{x_j}^2(1-\kappa))$. The soft-thresholding operator then applies component-wise, and the probability that predictor j is selected becomes

$$\mathbb{P}(|\beta_j^{\mathrm{OLS}}| > \lambda) = 1 - \left[\Phi\left(\frac{\lambda}{\sigma_{\mathrm{OLS},j}} \right) - \Phi\left(\frac{-\lambda}{\sigma_{\mathrm{OLS},j}} \right) \right], \tag{32}$$

where $\sigma_{\mathrm{OLS},j}^2 = \gamma\sigma_d^2/(2\sigma_{x_j}^2(1-\kappa))$. Since the β_j are mutually independent when predictors are uncorrelated, the expected number of selected predictors is simply $\mathbb{E}[\#\,\text{selected}] = \sum_{j=1}^k \mathbb{P}(|\beta_j^{\mathrm{OLS}}| > \lambda)$. This decomposition also extends to the stability analysis: the Jacobian of the mean dynamics becomes block-diagonal, with each block corresponding to a single predictor, and the SRA framework applies after vectorizing the parameter space as $\theta_t = (\beta_t, \mathrm{vec}(M_t))$. In summary, uncorrelated predictors preserve the key insight that sparse predictability arises from finite-sample fluctuations in individual coefficients, with each predictor independently subject to selection when its estimated relationship happens to exceed the penalty threshold.

7 Empirical Application

7.1 Data

This section evaluates the model's predictions in the data. Our sample consists of the cross-section of NYSE-listed stocks from 2010 to 2017. As predictors, we use the 180 daily news-topic attention series constructed by Bybee et al. (2021) from the full text of the *Wall Street Journal* using an unsupervised Latent Dirichlet Allocation (LDA) model. The LDA algorithm assigns each article a vector of topic proportions, which Bybee et al. (2021) aggregate to the daily level to measure media attention across a broad range of business, economic, political, and social topics. This high-dimensional predictor set is well suited to our framework, reflecting an environment in which agents observe many potential signals but have no prior knowledge about which of them are informative for returns.

In the empirical analysis, we use innovations in topic attention rather than levels. Media attention is highly persistent, so levels largely capture slow-moving components of news coverage. To isolate unexpected changes in information, we estimate an AR(1) process for each topic-attention series and define X_t as the corresponding residuals. The dependent variable is the logarithmic return of each stock. We observe all series at the daily frequency. Table 1 reports summary statistics.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	N
Log return	0.000	0.021	−1.964	2.051	−0.587	101.917	3,060,180
Topic level	0.006	0.004	0.000	0.116	3.662	33.361	340,020
AR(1) coefficient	−0.021	0.039	−0.277	0.007	−3.641	16.670	180
Topic innovation	0.000	0.003	−0.054	0.062	1.885	16.019	339,840

Note: The table reports summary statistics for daily log returns $r_{i,t} = \log(1 + R_{i,t})$ of NYSE-listed stocks from 2010–2017 and for news-topic attention measures constructed following Bybee et al. (2021). Topic attention innovations are defined as residuals from topic-specific AR(1) regressions estimated over the full sample.

7.2 Estimation Strategy

Our empirical analysis is organized around a simple parallel: as econometricians, we face a learning and specification problem that is closely analogous to the one confronting agents in the model. In the theory, log returns are generated by the ALM, but neither agents nor the econometrician observe the true data-generating process. Accordingly, we distinguish between (i) the estimation problem solved by agents in real time and (ii) the ex-post estimation problem we solve using realized returns and predictors. As in the model, we endow agents with the following linear PLM for log returns:

$$r_{t+1} = X_t \beta_t + \eta^d_{t+1}, \tag{33}$$

where X_t is the matrix of topic-attention innovations. The closed-form LASSO results require orthonormal regressors. In the empirical counterpart, we adopt the weaker condition that predictors are approximately uncorrelated, so that equation (28) need not hold deterministically, but can be expected to provide a close approximation. Figure 2 in Appendix B.4 shows that the topic-attention innovations are close to uncorrelated contemporaneously, and Table 2 shows that cross-lag correlations are also small in the time series.

We further assume that agents estimate the PLM using a rolling window rather than geometrically declining weights. Specifically, at each date t the regressor matrix contains the past l observations:

$$X_t = \begin{bmatrix} x_{t-l} \\ x_{t-l+1} \\ \vdots \\ x_{t-1} \end{bmatrix}, \quad r_{t+1} = \begin{bmatrix} r_{t-l+1} \\ r_{t-l+2} \\ \vdots \\ r_t \end{bmatrix}. \tag{34}$$

For each rolling window, the agent computes the corresponding LASSO estimator β_t. Given this estimator, the ALM implies the modeler's PLM,

$$r_{t+1} = \log(1 - \kappa e^{X_t \beta_t}) - \log(1 - \kappa e^{X_{t+1}\beta_{t+1}}) + u_{t+1}. \tag{35}$$

Equation (35) suggests a direct empirical implementation: estimate the rolling-window LASSO that agents are assumed to run, use the resulting coefficients to construct the model-implied return component, and then estimate (35) by Non-Linear Least Squares (NLS) on the realizations of r_{t+1} and X_t. However, as discussed in the theoretical section, estimating a linear model in the data does not generally recover the primitive parameters used by agents. Instead, it identifies the transformed objects implied by the T-map. This observation motivates a three-step empirical strategy. First, estimate the (mis-specified) linear model using exactly the same specification we endow agents with. Second, invert the T-map to recover the parameters that agents would be using in their PLM. Finally, use these recovered parameters to estimate the NLS model implied by the ALM.

Before implementing our estimation strategy, we calibrate the three parameters governing the agent's learning rule: the rolling window length l, the penalty parameter λ, and the number of lags used for forecasting n. We select these parameters using Bayesian optimization as implemented in the Optuna package (**?**). Bayesian optimization constructs a probabilistic surrogate of the objective function and iteratively guides the search toward promising regions of the parameter space, allowing us to identify economically relevant configurations with relatively few function evaluations, which is an important consideration given the computational cost of our two-stage estimation procedure. The objective function maximizes the in-sample R^2 from the second-stage nonlinear least squares estimation. We penalize parameter combinations that yield statistically insignificant estimates, as our goal is to identify learning behavior that is both economically meaningful and statistically robust. We conduct the calibration using the CRSP value-weighted market return including dividends as the forecasting target and the full set of topic-innovation series as potential signals. Calibrating on the aggregate market return ensures that the resulting parameters reflect average learning dynamics and serve as meaningful benchmarks for the subsequent cross-sectional analysis. We perform 500 optimization trials to explore the parameter space. The optimization yields the following parameter values: a rolling window length of $l = 256$ days, a penalty parameter of $\lambda = 0.0025$, and a single forecasting lag ($n = 1$.

7.3 Results

We are now ready to apply our estimation strategy to the cross-section of stock returns. First, we estimate the LASSO regression on rolling windows of historical data for each stock to recover the time series of belief parameters $\{\beta_t\}$. The results are summarized in Table **??**

In Figure 1 we plot the average number of topics selected per estimation and stock against the variance of each stock's return. We find that stocks with more volatile returns tend to have a higher fraction of non-zero coefficients. The figure aligns with the theoretical prediction from Proposition 3 that higher return volatility leads to more frequent selection of predictors.

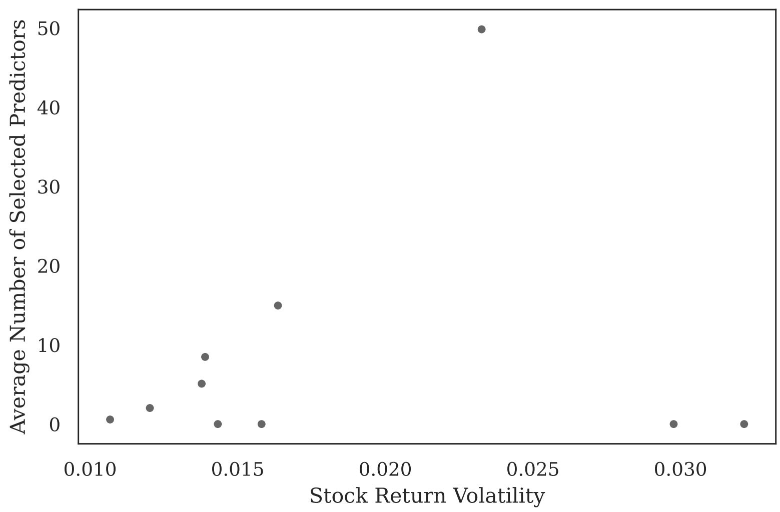

Figure 1: Average Number of Non-Zero Coefficients vs. Return Variance

Next, we feed the first-stage LASSO estimates into the ALM defined by equation (35) and estimate the structural parameter κ via Non-Linear Least Squares (NLS) for each stock. Table **??** shows the summary of this estimation for the cross-section of stocks.

Our model shows that...

8 Conclusion

References

Adam, Klaus, Albert Marcet, and Juan Pablo Nicolini. 2016. "Stock Market Volatility and Learning." *Journal of Finance*, 71.

Adam, Klaus, and Albert Marcet. 2011. "Internal rationality, imperfect market knowledge and asset prices." *Journal of Economic Theory*, 146(3): 1224–1252.

Barberis, Nicholas, Andrei Shleifer, and Robert Vishny. 1998. "A model of investor sentiment." *Journal of Financial Economics*, 49(3): 307–343.

Begenau, Juliane, Maryam Farboodi, and Laura Veldkamp. 2018. "Big data in finance and the growth of large firms." *Journal of Monetary Economics*, 97: 71–87.

Branch, William A., and George W. Evans. 2010. "Asset return dynamics and learning." *Review of Financial Studies*, 23.

Bybee, Leland, Bryan T. Kelly, Asaf Manela, and Dacheng Xiu. 2021. "Business News and Business Cycles." *SSRN Electronic Journal*.

Chinco, Alex, Adam D. Clark-Joseph, and Mao Ye. 2019. "Sparse Signals in the Cross-Section of Returns." *Journal of Finance*, 74.

Evans, George W., and Seppo Honkapohja. 2001. *Learning and Expectations in Macroeconomics.* Princeton University Press.

Farboodi, Maryam, and Laura Veldkamp. 2020. "Long-run growth of financial data technology." *American Economic Review*, 110(8): 2485–2523.

Farmer, Leland E., Lawrence Schmidt, and Allan Timmermann. 2023. "Pockets of Predictability." *Journal of Finance*, 78.

Feng, Guanhao, Stefano Giglio, and Dacheng Xiu. 2020. "Taming the Factor Zoo: A Test of New Factors." *Journal of Finance*, 75.

Freyberger, Joachim, Andreas Neuhierl, and Michael Weber. 2020. "Dissecting characteristics nonparametrically." *The Review of Financial Studies*, 33(5): 2326–2377.

Georges, Christophre, and Javier Pereira. 2021. "Market stability with machine learning agents." *Journal of Economic Dynamics and Control*, 122.

Gu, Shihao, Bryan Kelly, and Dacheng Xiu. 2020. "Empirical Asset Pricing via Machine Learning." *Review of Financial Studies*, 33.

Hommes, Cars, and Mei Zhu. 2014. "Behavioral learning equilibria." *Journal of Economic Theory*, 150.

Hong, Harrison, Jeremy C Stein, and Jialin Yu. 2007. "Simple Forecasts and Paradigm Shifts." *The Journal of Finance*, 62(3): 1207–1242.

Honkapohja, Seppo, and Kaushik Mitra. 2003. "Learning with bounded memory in stochastic models." *Journal of Economic Dynamics and Control*, 27.

Invesco. 2024. "Invesco Global Systematic Investing Study 2024." Invesco. Accessed: 09-12-2025.

Kozak, Serhiy, Stefan Nagel, and Shrihari Santosh. 2020. "Shrinking the cross-section." *Journal of Financial Economics*, 135(2): 271–292.

Lucas, Robert E. 1978. "Asset Prices in an Exchange Economy." *Econometrica*, 46.

Marcet, Albert, and Thomas J. Sargent. 1989. "Convergence of least squares learning mechanisms in self-referential linear stochastic models." *Journal of Economic Theory*, 48.

Mclean, R. David, and Jeffrey Pontiff. 2016. "Does Academic Research Destroy Stock Return Predictability?" *Journal of Finance*, 71.

Tibshirani, Robert. 1996. "Regression Shrinkage and Selection Via the Lasso." *Journal of the Royal Statistical Society. Series B: Methodological*, 58.

Timmermann, Allan. 1996. "Excess Volatility and Predictability of Stock Prices in Autoregressive Dividend Models with Learning." *Review of Economic Studies*, 63.

Timmermann, Allan G. 1993. "How learning in financial markets generates excess volatility and predictability in stock prices." *Quarterly Journal of Economics*, 108.

Wright, Adam. 2025. "Worldwide IDC Global DataSphere Forecast, 2025–2029." International Data Corporation (IDC) Market Forecast Report US53363625. Accessed: ¡insert date you accessed the report¿.

A Derivation of Recursive Learning Models

A.1 Recursive Least Squares

The first updating mechanism we consider is Recursive Least Squares (RLS), which is a recursive implementation of the standard Ordinary Least Squares (OLS). In OLS, the agent estimates the coefficients of a linear regression model by minimizing the sum of squared residuals. Assuming agents use OLS to estimate the parameters in the PLM (10), the OLS estimator β_t can be derived from the minimization[4] problem as

$$\beta_t = \left(\sum_{i=1}^{t-1} x_{i-1}^2 \right)^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{36}$$

[4]We use the following timing: at time t the agent forecasts $r_{t+1}^e = x_t \beta_t$ which then implies a realization for r_t. Therefore, to avoid simultaneity issues, we assume that when computing the current estimate, the last available log-return observation is r_{t-1}.

Estimating OLS on the whole available data set is sometimes referred to as off-line estimation. Another approach is to estimate the model on-line, that is, as soon as a new data point arrives. It turns out that the estimator formula allows for a recursive formulation. Define

$$M_{t-1} = \frac{1}{t-1}\left(\sum_{i=1}^{t-1} x_{i-1}^2\right), \quad \beta_t = \frac{1}{t-1} M_{t-1}^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{37}$$

First we establish a recursion for M_t. Start from

$$\frac{t-1}{t} M_{t-1} = \frac{1}{t}\left(\sum_{i=1}^{t-1} x_{i-1}^2\right) \implies \frac{t-1}{t} M_{t-1} + \frac{1}{t} x_{t-1}^2 = M_t, \tag{38}$$

that is

$$M_t = M_{t-1} + \frac{1}{t}\left(x_{t-1}^2 - M_{t-1}\right). \tag{39}$$

For β_t we have

$$\begin{aligned} \beta_t &= \beta_{t-1} + [(t-1)M_{t-1}]^{-1}\left\{x_{t-2} r_{t-1} + [(t-2)M_{t-2}]\,\beta_{t-1}\right\} - \beta_{t-1} \\ &= \beta_{t-1} + [(t-1)M_{t-1}]^{-1}\left\{x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-1}\right\}. \end{aligned} \tag{40}$$

A.2 Constant Gain Learning

In Weighted Least Squares (WLS), the agent estimates the coefficients of a linear regression model by minimizing a weighted sum of squared residuals. The weights are typically chosen to reflect the relative importance or reliability of different observations. Assuming the linear model (10), the WLS estimator can be derived from the minimization problem

$$\min_{\beta} \sum_{i=1}^{t-1} w_{t-1,i}(r_i - \beta x_{i-1})^2, \tag{41}$$

with solution

$$\beta = \left(\sum_{i=1}^{t-1} w_{t-1,i} x_{i-1}^2\right)^{-1} \sum_{i=1}^{t-1} w_{t-1,i} x_{i-1} r_i. \tag{42}$$

In particular, we are interested in the case where the weights decline geometrically over time[5]

$$w_{t,i} = \gamma(1-\gamma)^{t-i}, \quad \gamma \in (0,1). \tag{43}$$

Similar to the OLS case, a recursive relationship can be derived.

[5]In empirical applications, instead of using geometrically declining weights, it is common to use a rolling window approach, where only the most recent observations within a fixed-size window are used for estimation. However, the latter is analytically intractable. It is possible to determine a relationship between the window size and the geometric decay parameter γ such that the two methods give similar weights *overall* on the ℓ most recent observations: $\ell \approx \log(1-p)/\log(1-\gamma)$, where p is the proportion of total weight assigned to the ℓ most recent observations.

Define

$$M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1}^2, \quad \beta_t = \gamma M_{t-1}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i. \tag{44}$$

First we establish a recursion for M_t. Start from

$$M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1}^2, \tag{45}$$

then

$$(1-\gamma) M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-i} x_{i-1}^2 = M_t - \gamma x_{t-1}^2. \tag{46}$$

That is

$$M_t = M_{t-1} + \gamma \left(x_{t-1}^2 - M_{t-1} \right). \tag{47}$$

Similarly for β_t we start from the lagged expression

$$\beta_{t-1} = \gamma M_{t-2}^{-1} \sum_{i=1}^{t-2} (1-\gamma)^{t-2-i} x_{i-1} r_i, \tag{48}$$

then

$$(1-\gamma)\beta_{t-1} = \gamma M_{t-2}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i, \tag{49}$$

and

$$\beta_{t-1}(M_{t-1} - \gamma x_{t-2}^2) = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i \rightarrow \beta_{t-1}(1 - \gamma M_{t-1}^{-1} x_{t-2}^2) = \gamma M_{t-1}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i, \tag{50}$$

since

$$M_{t-2} = \frac{1}{1-\gamma}(M_{t-1} - \gamma x_{t-2}^2).$$

Adding $\gamma M_{t-1}^{-1} x_{t-2} r_{t-1}$ to both sides we obtain

$$\beta_{t-1}(1 - \gamma M_{t-1}^{-1} x_{t-2}^2) + \gamma M_{t-1}^{-1} x_{t-2} r_{t-1} = \beta_t, \tag{51}$$

which can be rearranged as

$$\beta_t = \beta_{t-1} + \gamma M_{t-1}^{-1} x_{t-2} (r_{t-1} - x_{t-2} \beta_{t-1}). \tag{52}$$

A.3 Lasso Learning

The LASSO (Least Absolute Shrinkage and Selection Operator) estimator is defined as the solution to the optimization problem

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \frac{1}{2} \left[(Y - X\beta)'(Y - X\beta) \right] + \lambda \sum_{j=1}^{k} |\beta_j| \right\},$$

where $\lambda \geq 0$ is a tuning parameter that controls the amount of regularization. The LASSO estimator can be interpreted as a trade-off between fitting the data well (minimizing the sum of squared residuals) and keeping the model simple (minimizing the absolute values of the coefficients).

In general there is no closed form solution for the LASSO estimator, since the absolute value function is not differentiable at zero. However, in the special case where X is an orthonormal matrix, the LASSO estimator can be expressed in closed form. To do so we can rewrite the optimization problem as

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \Big\{ \frac{1}{2} \left[Y'Y - 2\beta' \underbrace{X'Y}_{\beta_{\text{OLS}}} + \beta' \underbrace{X'X}_{I} \beta \right] + \lambda \sum_{j=1}^{k} |\beta_j| \Big\},$$

where the underbrackets follow from orthonormality and notice that the term $Y'Y$ does not depend on β, so it does not change the minimizer. Using $A'A = \sum_{j=1}^{k} a_j^2$, we can rewrite the problem as

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \Big\{ \sum_{j=1}^{k} \Big(-\beta_j \beta_{\text{OLS},j} + \frac{1}{2}\beta_j^2 + \lambda|\beta_j| \Big) \Big\}.$$

This problem is separable in the coefficients β_j, so we can solve for each coefficient independently. The optimization problem for each j coefficient is

$$\min_{\beta_j} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda|\beta_j| \right\}.$$

To solve this problem, we consider two cases:

1. $\beta_{\text{OLS},j} > 0$, then β_j has to be positive, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

   $$\min_{\beta_j \geq 0} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda \beta_j \right\},$$

   and differentiating and setting to zero gives

   $$\beta_j - \beta_{\text{OLS},j} + \lambda = 0 \Rightarrow \beta_j = \beta_{\text{OLS},j} - \lambda.$$

Imposing non-negativity yields

$$\hat{\beta}_{\text{LASSO},j} = \max\left(0, \beta_{\text{OLS},j} - \lambda\right).$$

2. $\beta_{\text{OLS},j} < 0$, then β_j has to be negative, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

$$\min_{\beta_j < 0} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda(-\beta_j) \right\},$$

and

$$\beta_j - \beta_{\text{OLS},j} - \lambda = 0 \Rightarrow \beta_j = \beta_{\text{OLS},j} + \lambda.$$

Imposing negativity yields

$$\hat{\beta}_{\text{LASSO},j} = \min\left(0, \beta_{\text{OLS},j} + \lambda\right).$$

Combing the two cases, we obtain

$$\hat{\beta}_{\text{LASSO}} = S(\beta_{\text{OLS}}, \lambda) = \text{sign}(\beta_{\text{OLS}}) \cdot \max\left(0, |\beta_{\text{OLS}}| - \lambda\right), \tag{53}$$

where $S(\cdot)$ is the soft-thresholding operator.

B Proofs

B.1 Proof of proposition 1

In a CMCE the value of β solves

$$T(\beta) - \beta = 0, \tag{54}$$

where

$$T(\beta) = \frac{\text{Cov}(r_{t-1}, x_{t-2})}{\text{Var}(x_{t-2})} = \frac{\text{Cov}(r_{t-1}, x_{t-2})}{\sigma_x^2}, \tag{55}$$

and r_{t-1} is given by (23) evaluated at the stationary distribution.

To compute $\text{Cov}(r_{t-1}, x_{t-2})$ we exploit the fact that x_{t-2} is independent of η_{t-1}^d and x_{t-1}, which implies that

$$\text{Cov}(r_{t-1}, x_{t-2}) = \text{Cov}\left(g_\beta(x_{t-2}), x_{t-2}\right). \tag{56}$$

By Stein's lemma

$$\text{Cov}(r_{t-1}, x_{t-2}) = \text{Cov}(g_\beta(x_{t-2}), x_{t-2}) = \sigma_x^2 \mathbb{E}\left[g'_\beta(x_{t-2})\right]. \tag{57}$$

This implies that $T(\beta) = \mathbb{E}\left[g'_\beta(x_{t-2})\right]$.

We can show that $\text{sign}\,[T(\beta)] = -\,\text{sign}(\beta)$ unless $\beta = 0$. In fact, we have

$$g'_\beta(x) = -\frac{\kappa\,\beta\,e^{w\tanh(x\beta/w)}\,\text{sech}^2(x\beta/w)}{1-\kappa e^{w\tanh(x\beta/w)}}. \tag{58}$$

In the numerator, every multiplicative factor except β is strictly positive since $\kappa > 0$, $e^{w\tanh(x\beta/w)} > 0$, and $\text{sech}^2(x\beta/w) > 0$. The denominator is also strictly positive since w is chosen exactly to make the second term smaller than 1. Hence the sign of $g'_\beta(x)$ is determined entirely by $-\beta$. In particular,

$$\text{sign}\,[T(\beta)] = \text{sign}\left[-\beta\,\mathbb{E}\left(\frac{\kappa\,e^{w\tanh(x\beta/w)}\,\text{sech}^2(x\beta/w)}{1-\kappa e^{w\tanh(x\beta/w)}}\right)\right] = -\,\text{sign}(\beta). \tag{59}$$

Therefore $T(\beta)$ has the opposite sign of β for all $\beta \neq 0$, and the only fixed point is $\beta = 0$.

To study local stability of the CMCE under RLS learning we follow the approach in Chapter 6 of Evans and Honkapohja (2001), which casts the learning dynamics as a Stochastic Recursion Algorithm (SRA). The key technical issue is that the return innovation r_{t-1} depends on both β_{t-1} and β_{t-2}. To deal with this, we keep the *parameter* vector first order and move the required lags into the *state* vector. Specifically, define the parameter vector as

$$\theta_t \equiv \begin{pmatrix} \beta_t \\ M_t \end{pmatrix},$$

with domain $D = \mathbb{R} \times (0, \infty)$, and the gain sequence $\gamma_t = t^{-1}$.

We augment the state to include the lagged belief needed by the ALM:

$$Z_t \equiv \begin{pmatrix} x_{t-1} \\ x_{t-2} \\ \eta_{t-1} \\ s_{t-1} \\ s_{t-2} \end{pmatrix}, \qquad s_{t-1} \equiv \beta_{t-1}.$$

Let $W_t = (x_{t-1}, \eta_{t-1}, 1)'$. Then the state follows a conditionally linear law of motion of the form

$$Z_t = AZ_{t-1} + B(\theta_{t-1})W_t, \quad A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad B(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \beta \\ 0 & 0 & 0 \end{pmatrix}.$$

This construction ensures $s_{t-1} = \beta_{t-1}$ in real time, while under a fixed parameter $\theta = (\beta, M)$ we have $s \equiv \beta$ in the associated frozen-parameter state process $\bar{Z}_t(\theta)$. Next, write the parameter recursion in SRA form as

$$\theta_t = \theta_{t-1} + \gamma_t\, \mathcal{H}(\theta_{t-1}, Z_t) + \gamma_t^2\, \rho_t(\theta_{t-1}, Z_t),$$

by rearranging (40) to get

$$\begin{aligned} \beta_t &= \beta_{t-1} + t^{-1} M_{t-1}^{-1} \left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-1} \right] \\ &\quad + t^{-2} \frac{t}{t-1} M_{t-1}^{-1} \left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-2} \right]. \end{aligned} \tag{60}$$

and β_{t-2} is replaced by the state component s_{t-2}, so that the required functions are

$$\mathcal{H}(\theta, Z) = \begin{pmatrix} M^{-1}[x_2\, r(\beta, s_2; x_1, x_2, \eta) - x_2^2\, \beta] \\ x_1^2 - M \end{pmatrix}, \qquad \rho_t(\theta, Z) = \begin{pmatrix} \frac{t}{t-1}\, M^{-1}[x_2\, r(\beta, s_2; x_1, x_2, \eta) - x_2^2\, s_2] \\ 0 \end{pmatrix},$$

with $Z = (x_1, x_2, \eta, s_1, s_2)$ and

$$r(\beta, s_2; x_1, x_2, \eta) = \eta + \log\Bigl(1 - \kappa e^{w \tanh(x_2 s_2/w)}\Bigr) - \log\Bigl(1 - \kappa e^{w \tanh(x_1 \beta/w)}\Bigr).$$

Assumptions (A.1)–(A.3) and (B.1)–(B.2) in Section 6.2.1 of Evans and Honkapohja (2001) hold on any compact $Q \subset D$: (A.1) holds because $\gamma_t = t^{-1}$ implies $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 < \infty$; (B.1) holds because W_t is i.i.d. with finite moments of all orders under $\eta_{t-1} \sim \mathcal{N}(0, \sigma^2)$ and $x_t \sim \mathcal{N}(0, \sigma_x^2)$; (B.2) holds since A is constant with spectral radius $0 < 1$ and $B(\theta)$ is uniformly bounded and Lipschitz on Q; (A.2) holds because $\tanh(\cdot)$ and $\operatorname{sech}(\cdot)$ are bounded and, under $\kappa e^w < 1$, the term $\log(1 - \kappa e^{w \tanh(\cdot)})$ is uniformly bounded away from its singularity on Q, implying polynomial growth of $\mathcal{H}$ and ρ_t in (x_1, x_2, ℓ), whose moments are finite; (A.3) follows because $\mathcal{H}$ is C^2 in θ on Q with bounded second derivatives (again using bounded derivatives of $\tanh$ and the restriction $\kappa e^w < 1$), hence satisfies the required Lipschitz properties. Therefore the associated ODE is

$$\dot{\theta} = h(\theta), \qquad h(\theta) = \lim_{t \to \infty} \mathbb{E}[\mathcal{H}(\theta, \bar{Z}_t(\theta))].$$

Under fixed $\theta = (\beta, M)$, the frozen state is stationary with $x_1, x_2 \overset{iid}{\sim} \mathcal{N}(0, \sigma_x^2)$, $\eta \sim \mathcal{N}(0, \sigma_d^2)$ and $s \equiv \beta$, so that

$$h(\beta, M) = \begin{pmatrix} M^{-1}(\mathbb{E}[x_2 r(\beta, \beta; x_1, x_2, \ell)] - \beta \sigma_x^2) \\ \sigma_x^2 - M \end{pmatrix} = \begin{pmatrix} (\sigma_x^2/M)(T(\beta) - \beta) \\ \sigma_x^2 - M \end{pmatrix},$$

where $\mathbb{E}[x_2 r(\beta, \beta; \cdot)] = \sigma_x^2 T(\beta)$ by the definition of $T(\beta)$. The CMCE corresponds to the unique fixed point $(\beta^*, M^*) = (0, \sigma_x^2)$. Linearizing h at $(0, \sigma_x^2)$ yields Jacobian

$$Dh(0, \sigma_x^2) = \begin{pmatrix} T'(0) - 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Moreover, from $g'_\beta(x) = -(\kappa \beta e^{w \tanh(x\beta/w)} \operatorname{sech}^2(x\beta/w))/(1 - \kappa e^{w \tanh(x\beta/w)})$ we obtain

$T(\beta) = \mathbb{E}[g'_\beta(x)]$ and hence $T'(0) = -(\kappa/(1-\kappa))$, so $T'(0) - 1 = -1/(1-\kappa) < 0$ for all $\kappa \in (0,1)$. Thus $(0, \sigma_x^2)$ is locally asymptotically stable for the ODE, and by the E-stability principle it is also locally stable under RLS learning.

B.2 Proof of proposition 2

For the proof we use the small-γ Gaussian approximation following Chapter 7 in Evans and Honkapohja (2001). We use the same state-augmentation device as above to eliminate the apparent second-order dependence, and all objects are defined as in the RLS case, except that now the gain sequence is constant: $\gamma_t \equiv \gamma \in (0,1)$. To verify the assumptions for the Gaussian approximation, we work on an open domain $D = \{(\beta, M) : \beta \in \mathbb{R},\ M \in (\zeta, \infty)\}$ with $\zeta > 0$ and (as standard in constant-gain RLS) impose a projection onto an arbitrary compact $Q \Subset D$ to guarantee $M_t \geq \zeta$; on Q, (A.2) and (A.3′) hold because $H(\theta, Z)$ is C^1 in (θ, Z) with polynomial growth in (x_1, x_2, η), and $\tanh(\cdot)$ together with $\kappa e^w < 1$ keep the logarithmic term uniformly away from its singularity, implying local Lipschitz and polynomial bounds for H and $\partial H/\partial x$; (M.1)–(M.5) and (N.1)(i)–(iii) hold because the augmented state is a geometrically ergodic Markov chain under fixed θ (linear shift with i.i.d. Gaussian innovations) and has finite moments of all orders; (H.1) holds since $h(\theta) \equiv \mathbb{E}[H(\theta, Z^\theta)]$ has continuous first and second derivatives on D; (H.2)–(H.3) hold locally around the CMCE because the Jacobian $B \equiv D_\theta h(\theta^*)$ has eigenvalues with strictly negative real parts (computed below), and $D_\theta h$ is Lipschitz on Q. Therefore Theorem 7.9 applies, yielding for small γ and large t the approximation $\theta_t \approx N(\theta^*, \gamma C)$ with

$$C = \int_0^\infty e^{sB} R(\theta^*) e^{sB'}\, ds, \qquad R_{ij}(\theta) = \sum_{k=-\infty}^{\infty} \mathrm{cov}(H_i(\theta, Z_k^\theta), H_j(\theta, Z_0^\theta)),$$

where Z_k^θ denotes the stationary frozen-θ chain. In our case, under frozen $\theta = (\beta, M)$ we have

$$h(\beta, M) = \begin{pmatrix} (\sigma_x^2/M)(T(\beta) - \beta) \\ \sigma_x^2 - M \end{pmatrix},$$

so the unique equilibrium is $\theta^* = (\beta^*, M^*) = (0, \sigma_x^2)$ and

$$B = D_\theta h(\theta^*) = \begin{pmatrix} T'(0) - 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{1-\kappa} & 0 \\ 0 & -1 \end{pmatrix},$$

where $T'(0) = -(\kappa/(1-\kappa))$ as in the decreasing-gain analysis. At θ^*, $r(\beta, s; \cdot) = \eta$ because $\tanh(0) = 0$ and the two log-terms cancel, hence

$$H_1(\theta^*, Z) = \sigma_x^{-2} x_2 \eta, \qquad H_2(\theta^*, Z) = x_1^2 - \sigma_x^2.$$

Because η has mean 0 and (x_t) and (η_t) are i.i.d. across time, all lead-lag covariances in $R(\theta^*)$ vanish and $R(\theta^*) = \mathrm{var}(H(\theta^*, Z))$ is diagonal with

$$R_{11}(\theta^*) = \mathrm{var}(\sigma_x^{-2} x_2 \eta) = \sigma_x^{-4}\,\mathbb{E}[x_2^2]\mathbb{E}[\eta^2] = \frac{\sigma_d^2}{\sigma_x^2}, \qquad R_{22}(\theta^*) = \mathrm{var}(x_1^2) = \mathbb{E}[x_1^4] - \sigma_x^4 = 2\sigma_x^4,$$

and $R_{12}(\theta^*) = 0$. Since B is diagonal, the integral for C is explicit:

$$C_{11} = \int_0^\infty e^{2B_{11}s} R_{11}\, ds = \frac{R_{11}}{-2B_{11}} = \frac{\sigma_d^2}{\sigma_x^2} \cdot \frac{1-\kappa}{2}, \qquad C_{22} = \int_0^\infty e^{2B_{22}s} R_{22}\, ds = \frac{R_{22}}{-2B_{22}} = \sigma_x^4, \qquad C_{12} = 0.$$

Consequently, for small γ and large t,

$$\theta_t = \begin{pmatrix} \beta_t \\ M_t \end{pmatrix} \approx N\left(\begin{pmatrix} 0 \\ \sigma_x^2 \end{pmatrix}, \gamma \begin{pmatrix} \frac{1-\kappa}{2} \frac{\sigma_d^2}{\sigma_x^2} & 0 \\ 0 & \sigma_x^4 \end{pmatrix} \right),$$

so in particular $\beta_t \approx N(0,\ \gamma\, \frac{1-\kappa}{2} \frac{\sigma_d^2}{\sigma_x^2})$ and $M_t \approx N(\sigma_x^2,\ \gamma\, \sigma_x^4)$.

B.3 Proof of proposition 3

Under soft-thresholded constant-gain learning, agents first compute the unpenalized weighted least squares estimate β_t^{OLS} exactly as in Section 5, then apply the soft-thresholding operator $S(\cdot, \lambda)$ before forming expectations. The key observation is that the updating rule for β_t^{OLS} remains unchanged—only the mapping from estimates to forecasts is modified. Formally, the learning dynamics are

$$\beta_t^{\mathrm{OLS}} = \beta_{t-1}^{\mathrm{OLS}} + \gamma M_{t-1}^{-1} x_{t-2}(r_{t-1} - x_{t-2}\beta_{t-1}^{\mathrm{OLS}}),$$

$$M_t = M_{t-1} + \gamma(x_{t-1}^2 - M_{t-1}),$$

and agents form forecasts using $\beta_t^{\mathrm{LASSO}} = S(\beta_t^{\mathrm{OLS}}, \lambda)$, which enters the actual law of motion as

$$r_t = \eta_t + \log\Big(1 - \kappa e^{w \tanh(x_{t-1} S(\beta_{t-1}^{\mathrm{OLS}}, \lambda)/w)}\Big) - \log\Big(1 - \kappa e^{w \tanh(x_t S(\beta_t^{\mathrm{OLS}}, \lambda)/w)}\Big).$$

The crucial insight is that β_t^{OLS} evolves according to the unpenalized system studied in Proposition 2. The soft-thresholding operator $S(\cdot, \lambda)$ affects prices and returns through the ALM, but the statistical properties of β_t^{OLS} itself are determined solely by the unpenalized updating rule. To verify that Proposition 2 applies, we check that all assumptions from the constant-gain analysis remain satisfied. The state-augmented system is unchanged, while the mapping H now includes the soft-thresholding operator through the return function $r(\cdot)$, but this does not affect regularity. Therfore it is enough to check assumptions (A.2) and (A.3′) for smoothness of H which hold because the soft-thresholding operator $S(\beta, \lambda)$ is Lipschitz continuous in β (with Lipschitz constant 1), and composition with the smooth $\tanh(\cdot)$ function preserves Lipschitz continuity of $r(\beta, s; \cdot)$ in (β, s). Since $\kappa e^w < 1$ by construction, the logarithmic terms remain uniformly bounded away from singularity. Thus H remains C^1 in θ with polynomial growth in Z.

The fixed point equations for the mean dynamics are now

$$\beta^{OLS*} = T(\beta^{L*}), \qquad \beta^{L*} = S(\beta^{OLS*}, \lambda).$$

In particular, $\beta^{OLS*} = 0$ implies $\beta^{L*} = 0$, so $(\beta^{OLS*}, M^*) = (0, \sigma_x^2)$ is always a fixed point. Any nonzero fixed point would require $|\beta^{OLS*}| > \lambda/\sigma_x^2$ and $\beta^{L*} = \beta^{OLS*} - \mathrm{sign}(\beta^{OLS*})\lambda$, implying

$$\beta^{OLS*} = -\frac{\kappa}{1-\kappa}\Big(\beta^{OLS*} - \mathrm{sign}(\beta^{OLS*})\lambda\Big) \;\Rightarrow\; \beta^{O*} = \mathrm{sign}(\beta^{O*})\,\frac{\kappa}{1+\kappa}\,\lambda,$$

which contradicts $|\beta^{O*}| > \lambda$ for all $\kappa \in (0,1)$. Hence for $\kappa \in (0,1)$ the unique fixed point is $(0, \sigma_x^2)$ and the corresponding LASSO coefficient is $\beta^{L*} = 0$. Local stability follows from the same argument as above, since $\beta^L = S(\beta^O, \lambda/M)$ equals 0 in a neighborhood of $\beta^{OLS} = 0$. Since $\beta_t^L = S(\beta_t^{OLS}, \lambda)$ is a deterministic transformation, the implied approximate stationary law of the LASSO coefficient is the *soft–thresholded Gaussian* and

$$\mathbb{P}(|\beta_{\text{OLS}}| > \lambda) = 1 - \left[\Phi\left(\frac{\lambda}{\sigma_{\text{OLS}}}\right) - \Phi\left(\frac{-\lambda}{\sigma_{\text{OLS}}}\right)\right],$$

where $\sigma_{\text{OLS}} = \left(\gamma s_d^2 \left(2\sigma_x^2\left(1 + \frac{\kappa}{1-\kappa}\right)\right)^{-1}\right)^{1/2}$.

B.4 Additional Tables and Figures

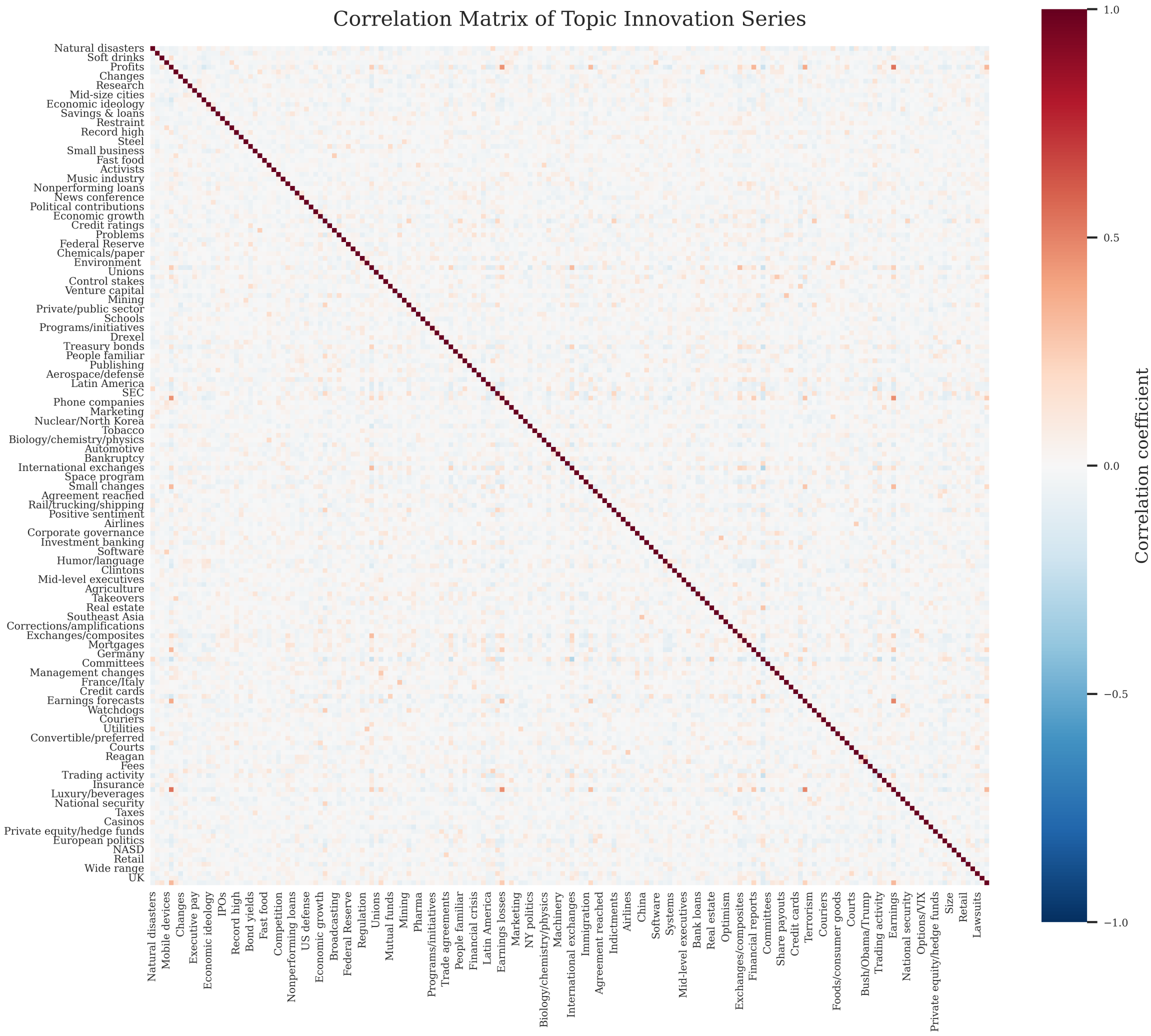

Figure 2: Correlation matrix of news-topic attention series.

Table 2: Cross-Lag Correlations of Predictors

Lag k	Lag l	Mean corr.	Median corr.	Max |corr.|	P95 |corr.|
1	1	−0.0003	−0.0037	0.5453	0.0904
1	3	0.0003	−0.0001	0.2147	0.0567
1	5	0.0002	−0.0002	0.1653	0.0572
1	7	0.0001	−0.0004	0.1592	0.0567
1	9	0.0002	0.0000	0.1514	0.0552
1	11	0.0004	0.0001	0.2656	0.0598
3	1	0.0003	−0.0001	0.2147	0.0567
3	3	−0.0003	−0.0037	0.5456	0.0904
3	5	0.0003	−0.0001	0.2146	0.0568
3	7	0.0002	−0.0001	0.1660	0.0573
3	9	0.0001	−0.0004	0.1594	0.0568
3	11	0.0002	−0.0001	0.1513	0.0552
5	1	0.0002	−0.0002	0.1653	0.0572
5	3	0.0003	−0.0001	0.2146	0.0568
5	5	−0.0003	−0.0037	0.5455	0.0903
5	7	0.0003	−0.0000	0.2116	0.0568
5	9	0.0002	−0.0001	0.1658	0.0572
5	11	0.0001	−0.0004	0.1603	0.0566
7	1	0.0001	−0.0004	0.1592	0.0567
7	3	0.0002	−0.0001	0.1660	0.0573
7	5	0.0003	−0.0000	0.2116	0.0568
7	7	−0.0002	−0.0038	0.5456	0.0902
7	9	0.0003	−0.0000	0.2115	0.0568
7	11	0.0002	−0.0002	0.1649	0.0572
9	1	0.0002	0.0000	0.1514	0.0552
9	3	0.0001	−0.0004	0.1594	0.0568
9	5	0.0002	−0.0001	0.1658	0.0572
9	7	0.0003	−0.0000	0.2115	0.0568
9	9	−0.0002	−0.0037	0.5454	0.0902
9	11	0.0003	−0.0000	0.2130	0.0567
11	1	0.0004	0.0001	0.2656	0.0598
11	3	0.0002	−0.0001	0.1513	0.0552
11	5	0.0001	−0.0004	0.1603	0.0566
11	7	0.0002	−0.0002	0.1649	0.0572
11	9	0.0003	−0.0000	0.2130	0.0567
11	11	−0.0002	−0.0036	0.5453	0.0904

Notes: The table reports pairwise correlations between predictors at lags k and l. For each lag pair, we report the mean and median correlation across predictors, the maximum absolute correlation, and the 95th percentile of the absolute correlation distribution. Diagonal entries ($k = l$) reflect same-lag correlations, while off-diagonal entries capture cross-lag dependence.